

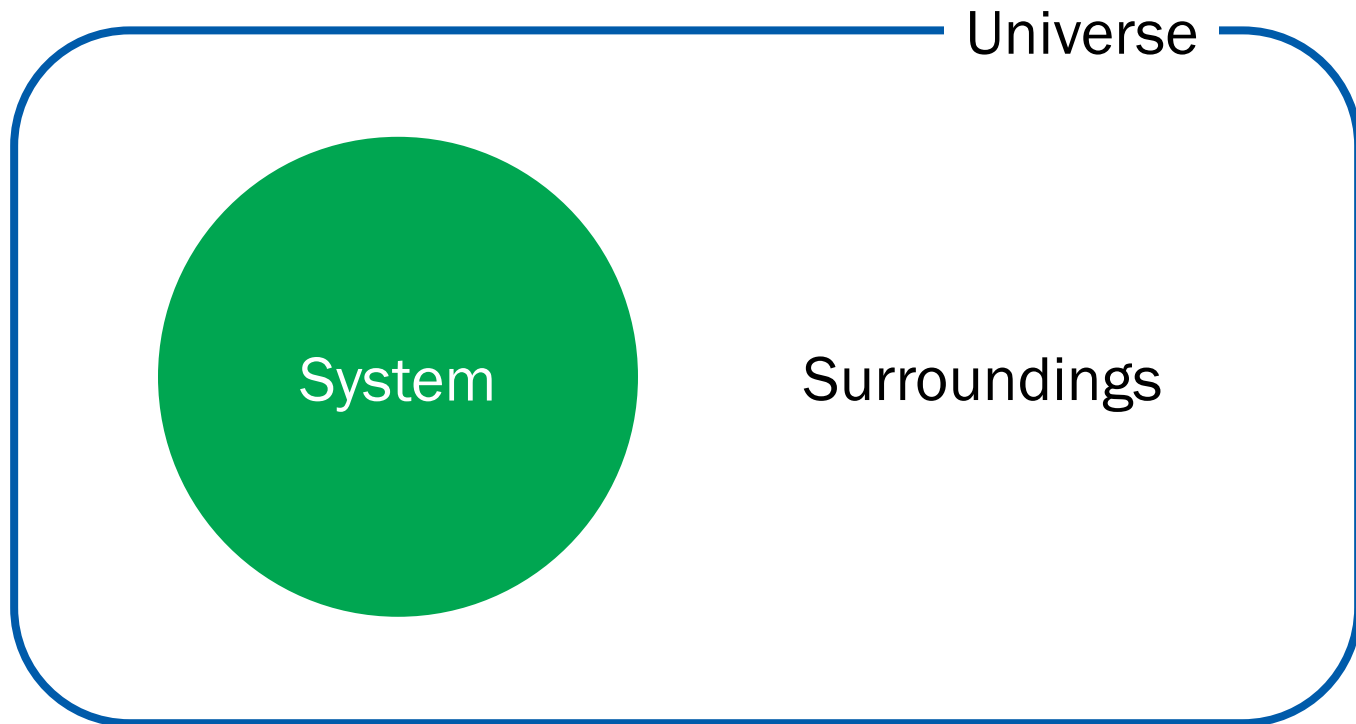
Physical Chemistry of Materials

Lecture 02

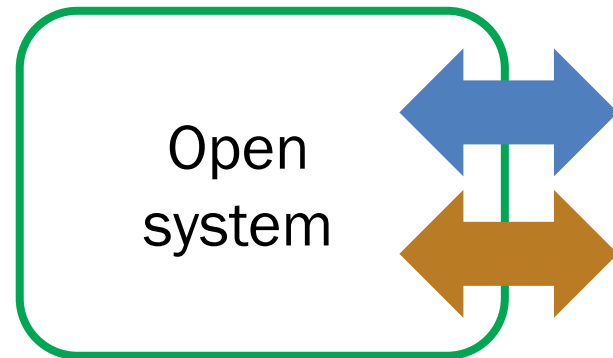
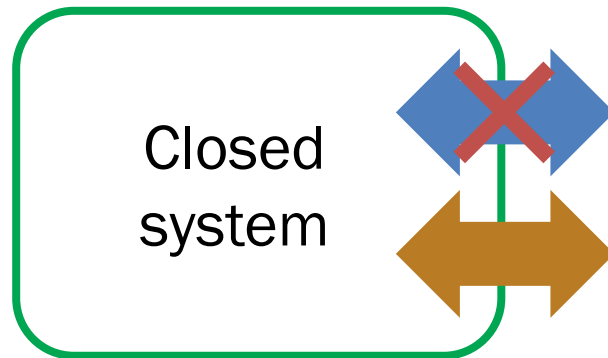
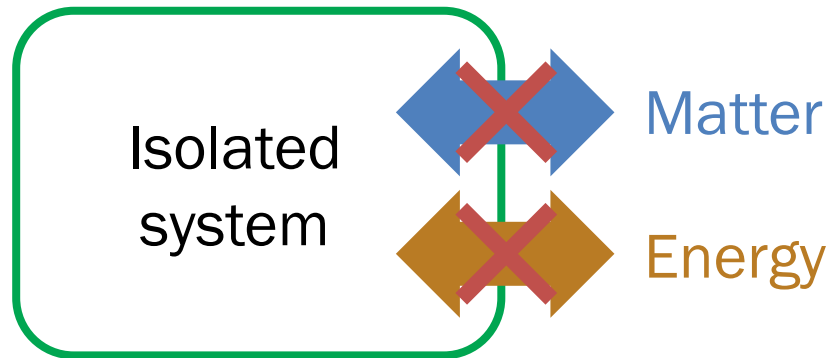
Basic Thermodynamics



System and Surroundings



Classification of Systems



System Properties

- A system can have different **physical properties**
- A **single value** will be assigned for each property
- Useful classification
 - **Extensive properties:** $f(A) + f(B) = f(A + B)$
 - **Intensive properties:** $f(A) + f(B) \neq f(A + B)$

Some Extensive Properties

- **Mass m :** measure of the quantity of matter present (unit: kilogram, kg)
- **Volume V :** measure of the quantity of space the sample occupies (unit: cubic meter, m³)
- **Amount of substance n :** measure of the number of specified entities (unit: mole, mol)

Some Intensive Properties

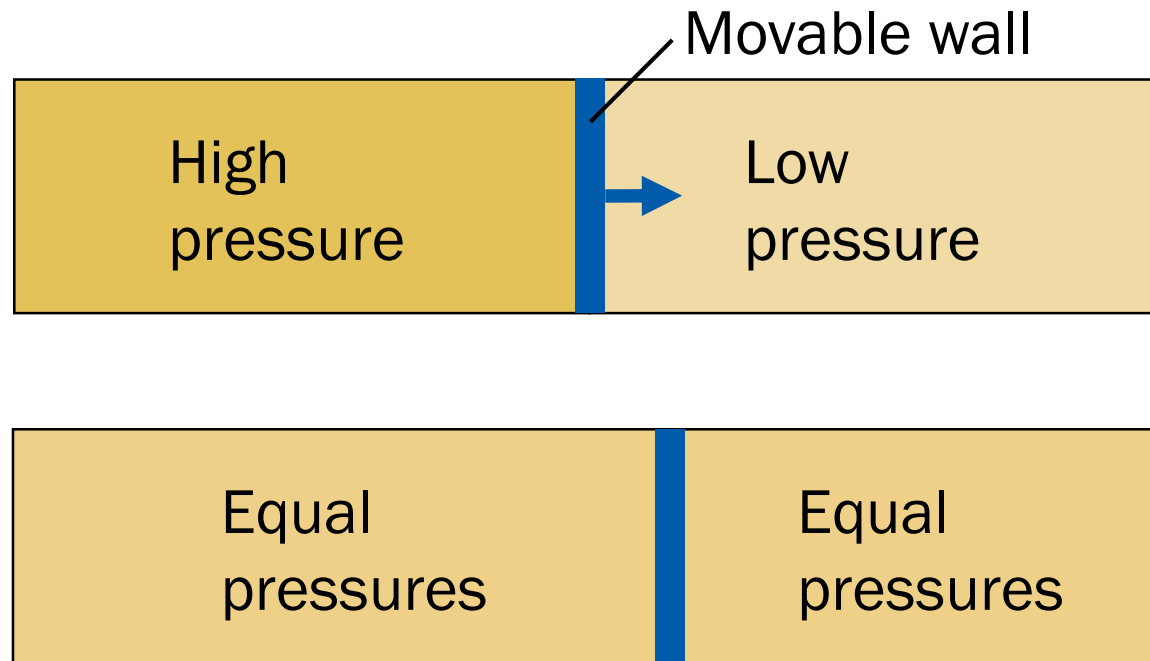
- **Pressure p :** force F / area A (unit: pascal, Pa)
 - Standard pressure $p^\ominus = 10^5 \text{ Pa} = 1 \text{ bar}$
 - Atmospheric pressure $101325 \text{ Pa} = 1 \text{ atm}$
- **Temperature T :** determines the direction of energy flow (unit: kelvin, K)
 - Celsius scale $\theta / ^\circ\text{C} = T / \text{K} - 273.15$

Equilibrium

- An isolated system is **in equilibrium** when its properties remain **constant with time**

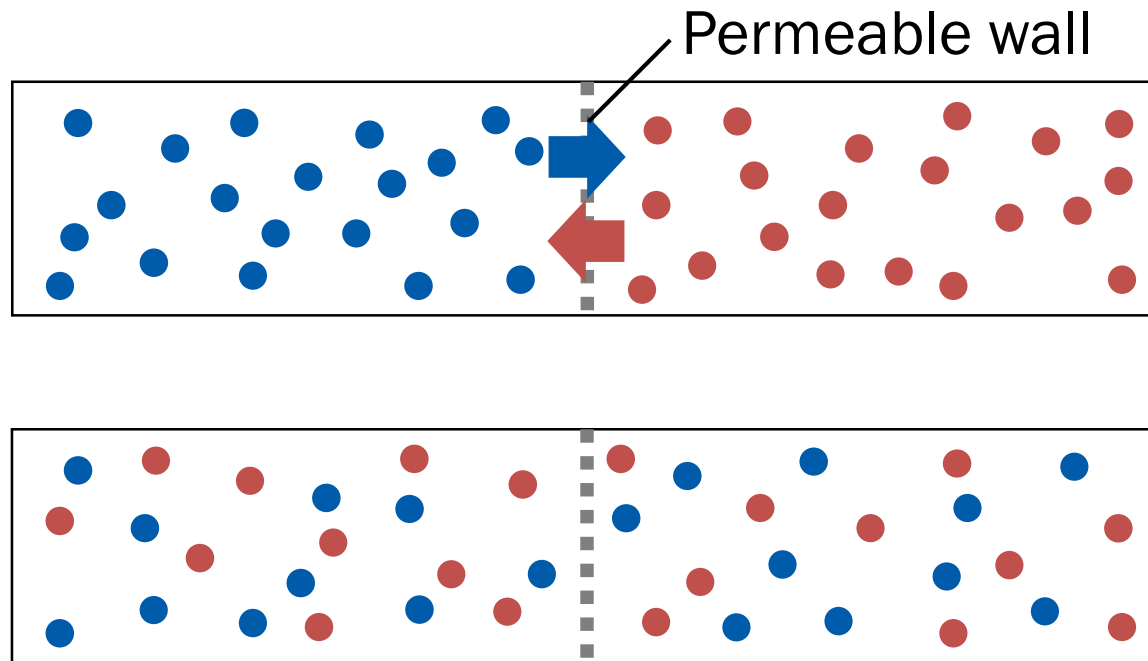
Mechanical Equilibrium

Constant p and V



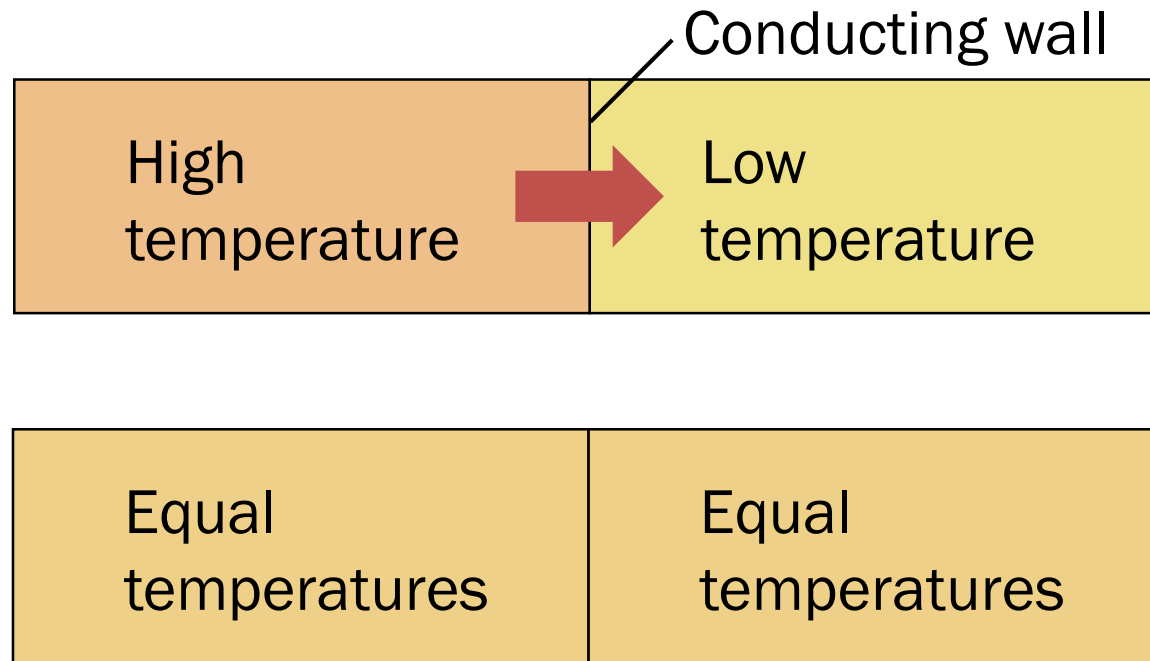
Diffusive Equilibrium

Constant n



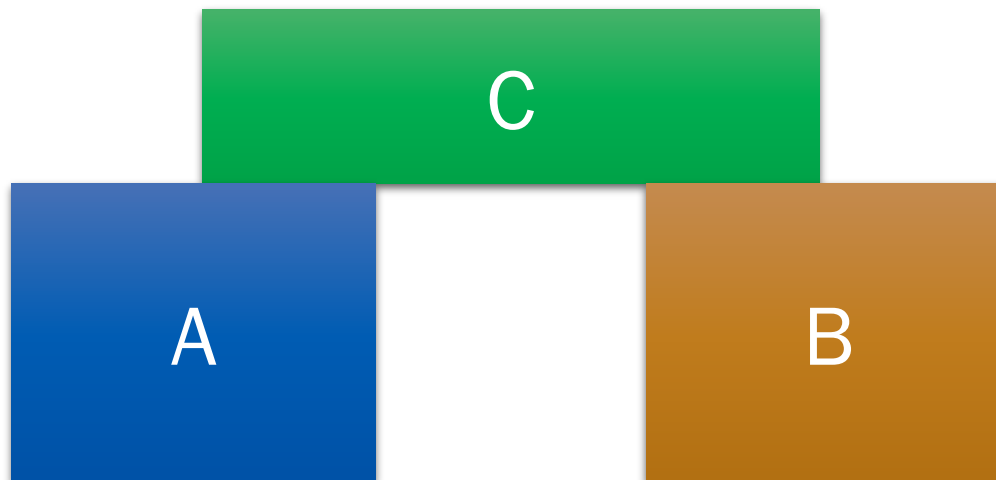
Thermal Equilibrium

Constant T



Zeroth Law of Thermodynamics

“If systems **A** and **C** are in thermal equilibrium, and systems **C** and **B** are in thermal equilibrium, systems **A** and **B** are in thermal equilibrium.”



State of a System

- The **state** of a system is defined by its **properties**
- Two samples of **the same substance** that have **the same properties** are in **the same state**
- Pure substance: (n, V, p, T)

State Change (Process)

The system can change its state
from one to another

$$(n_1, V_1, p_1, T_1) \rightarrow (n_2, V_2, p_2, T_2)$$

Equation of State

- The state is defined by (n, V, p, T)
- Equation of state: relationship of the 4 variables

$$p = f(T, V, n)$$

- Depends on a system (e.g. ideal gas)

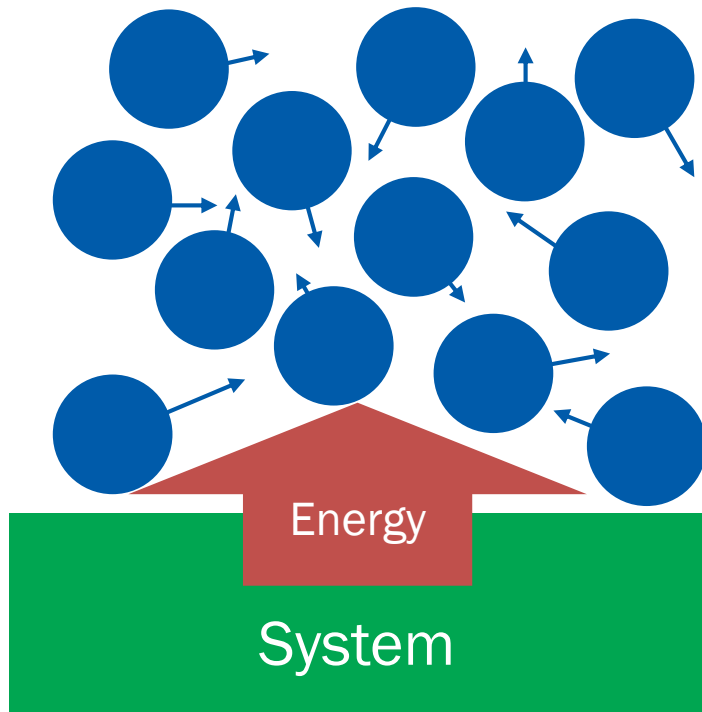
Internal Energy, U

“The total energy of a system”

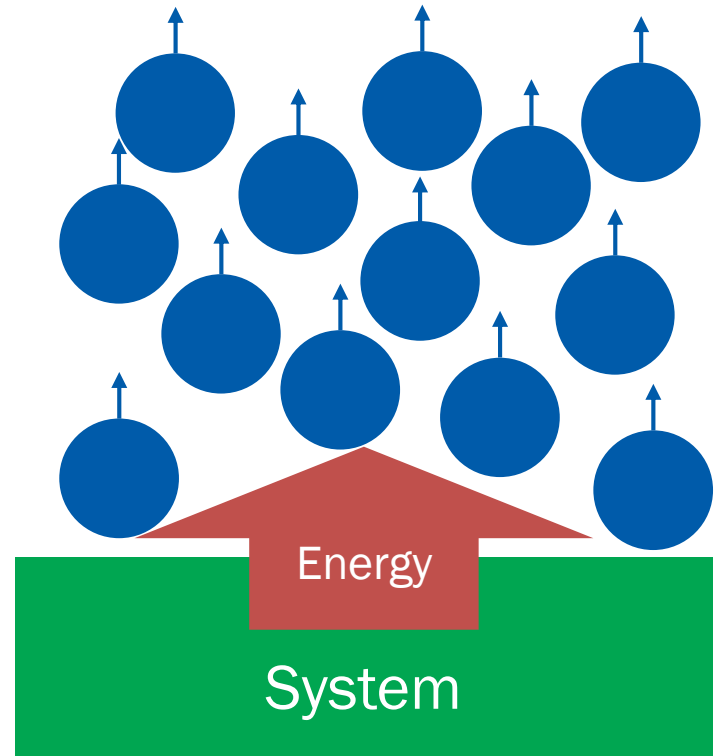
- Potential energy + kinetic energy of all particles
- Almost impossible to obtain absolute values

How can we change the internal energy?

Two Possible Ways



“Heat”



“Work”

First Law of Thermodynamics

“The internal energy of an isolated system is constant.”

- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$dU = \delta q + \delta w$$

↑ ↑ ↑
small change of U small q small w

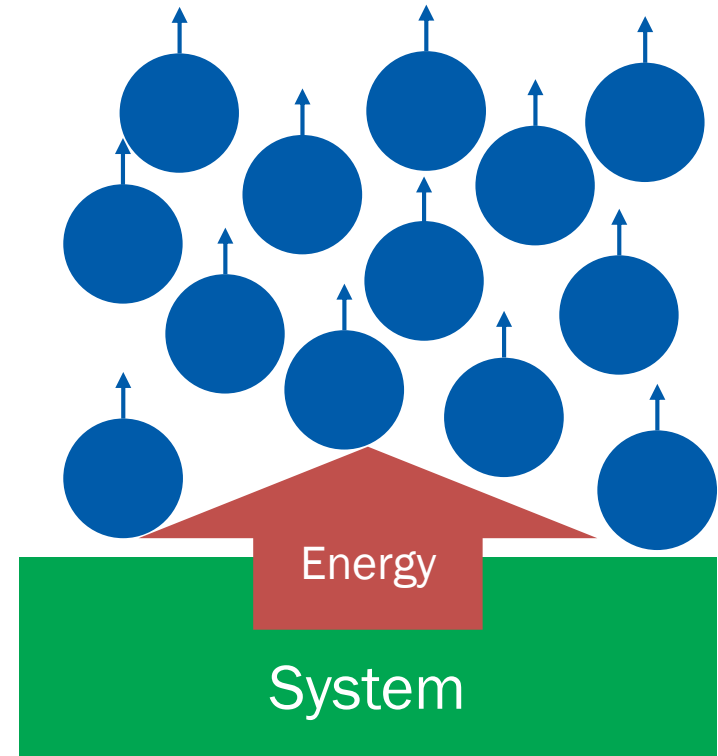
Work

$$\delta w = \vec{F} \cdot d\vec{s}$$

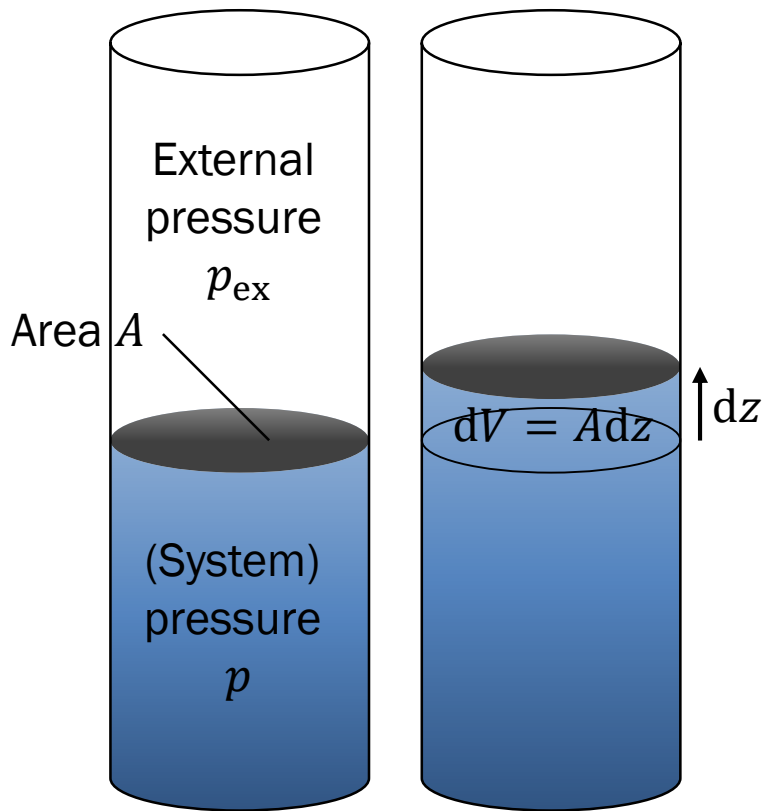
$\vec{F}$: “force” (intensive)

$d\vec{s}$: “displacement” (extensive)

- Sign indicates the direction of internal energy change



Expansion Work



$$\delta w = \vec{F} \cdot d\vec{s} = F_x dx + F_y dy + F_z dz$$

$$\text{As } dx = dy = 0,$$

$$\delta w = -|F|dz$$

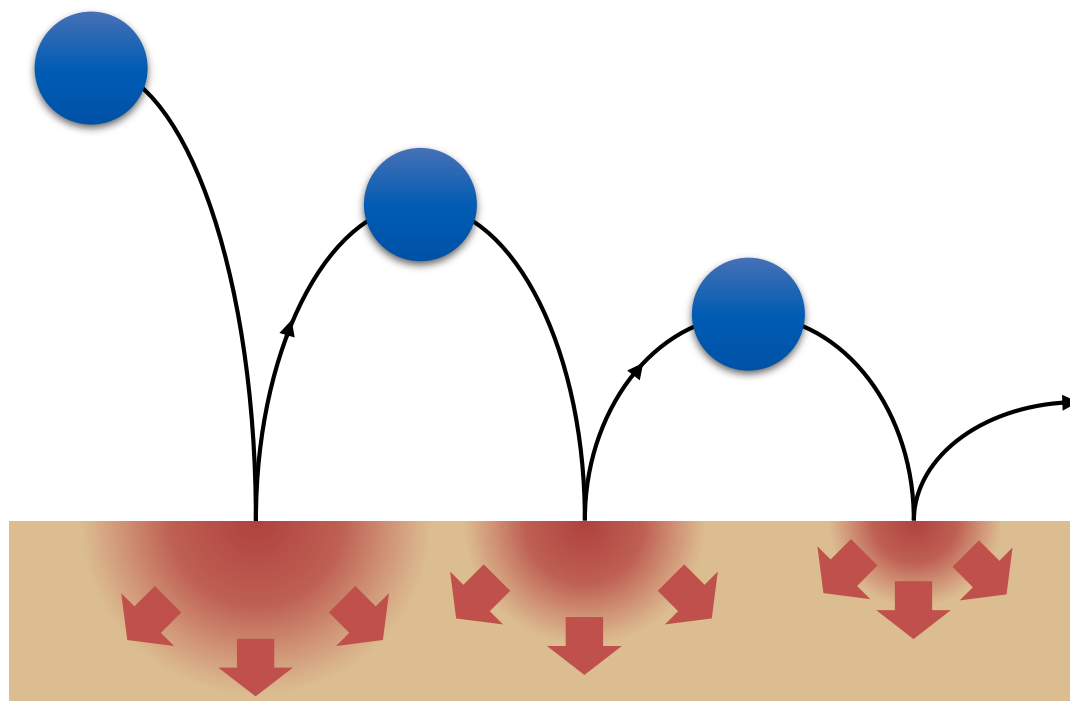
$$\text{Since } p_{\text{ex}} = |F|/A,$$

$$\delta w = -p_{\text{ex}}Adz = -p_{\text{ex}}dV$$

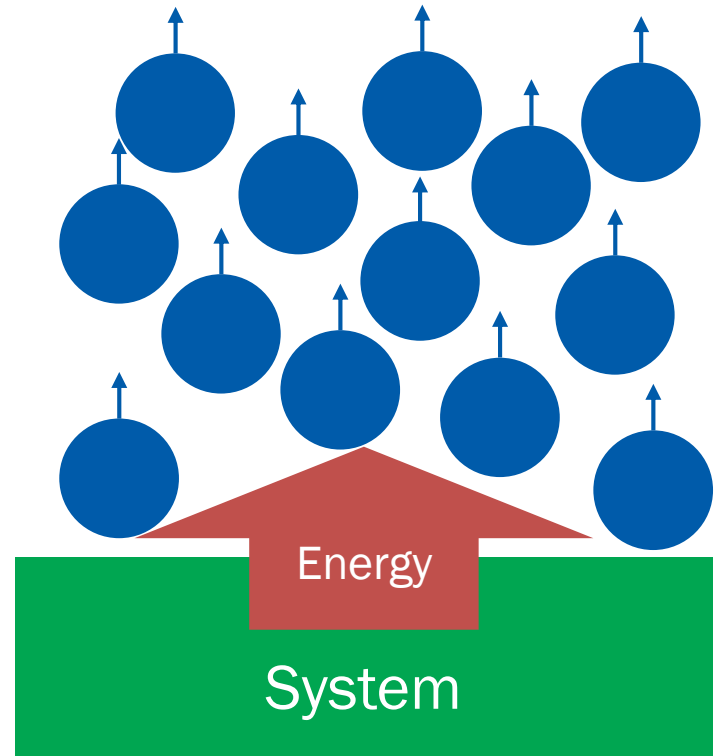
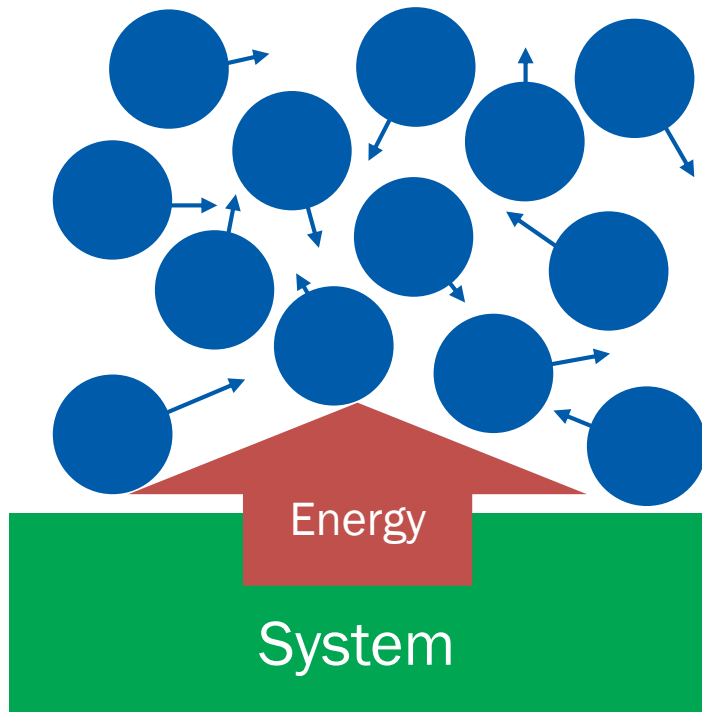
State and Path Functions

- **State function:** a property that depends only on the current state of a system and is independent of its history
e.g. internal energy, pressure, volume, temperature
- **Path function:** a property with a value that depends on the path between two states
e.g. heat, work

“Arrow of Time”



Why?



Heat and **work** are not equal!

Reversible vs. Irreversible Changes

- **Reversible change** consists of infinitesimal changes where the system is always **in equilibrium**
 - The reverse makes perfect sense
- **Irreversible change** involves **non-equilibrium** changes
 - The reverse looks strange
 - Arrow of time!

Entropy

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

- Index of exhaustion
- Second law of thermodynamics

$$dS \geq 0 \text{ (isolated system)}$$

- $dS = 0$: reversible change
- $dS > 0$: irreversible change

Third Law of Thermodynamics

The entropy of all perfect crystalline substances
is zero at $T = 0$

$$\lim_{T \rightarrow 0} S = 0$$

Clausius Inequality

$$dS_{\text{universe}} = dS_{\text{sys}} + dS_{\text{surr}} \geq 0$$

For the surroundings, every change is reversible;

$$dS_{\text{surr}} = \frac{\delta q_{\text{surr}}}{T}$$

Since $\delta q_{\text{surr}} = -\delta q_{\text{sys}}$,

$$dS_{\text{sys}} - \frac{\delta q_{\text{sys}}}{T} \geq 0$$

$$\therefore dS \geq \frac{\delta q}{T}$$

Fundamental Equation of Thermodynamics

From the first law,

$$dU = \delta q + \delta w$$

For a reversible change,

$$dU = \delta q_{\text{rev}} + \delta w_{\text{rev}} = TdS - pdV$$

Since all variables are state functions,

$$dU = TdS - pdV \text{ (in general)}$$

Natural Independent Variables

$$dU = TdS - pdV$$

→ S and V are **natural independent variables** of U

$$U = U(S, V)$$

As U is a state function,

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

Thermodynamic Definitions

$$dU = TdS - pdV$$

As $U = U(S, V)$ is a state function,

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

Comparison leads to

$$T = \left(\frac{\partial U}{\partial S} \right)_V$$

$$p = - \left(\frac{\partial U}{\partial V} \right)_S$$

Euler's Theorem

If $x = x(y, z)$ for extensive properties x , y , and z ,

$$x = \left(\frac{\partial x}{\partial y} \right)_z y + \left(\frac{\partial x}{\partial z} \right)_y z$$

Hence, for $U = U(S, V)$,

$$U = \left(\frac{\partial U}{\partial S} \right)_V S + \left(\frac{\partial U}{\partial V} \right)_S V = TS - pV$$

↑↑ ↑↑
conjugate pairs

Legendre Transforms (1)

$$U = U(S, V)$$

Can we get another function with different variables?

Subtract a conjugate pair!

- $(S, V) \rightarrow (S, p)$

$$H = U + pV \text{ (enthalpy)}$$

$$dH = dU + d(pV) = TdS + Vdp$$

$$\therefore H = H(S, p)$$

Legendre Transforms (2)

$$U = U(S, V)$$

Can we get another function with different variables?

Subtract a conjugate pair!

- $(S, V) \rightarrow (T, V)$

$$A = U - TS \text{ (Helmholtz energy)}$$

$$dA = dU - d(TS) = -SdT - pdV$$

$$\therefore A = A(T, V)$$

Legendre Transforms (3)

$$U = U(S, V)$$

Can we get another function with different variables?

Subtract a conjugate pair!

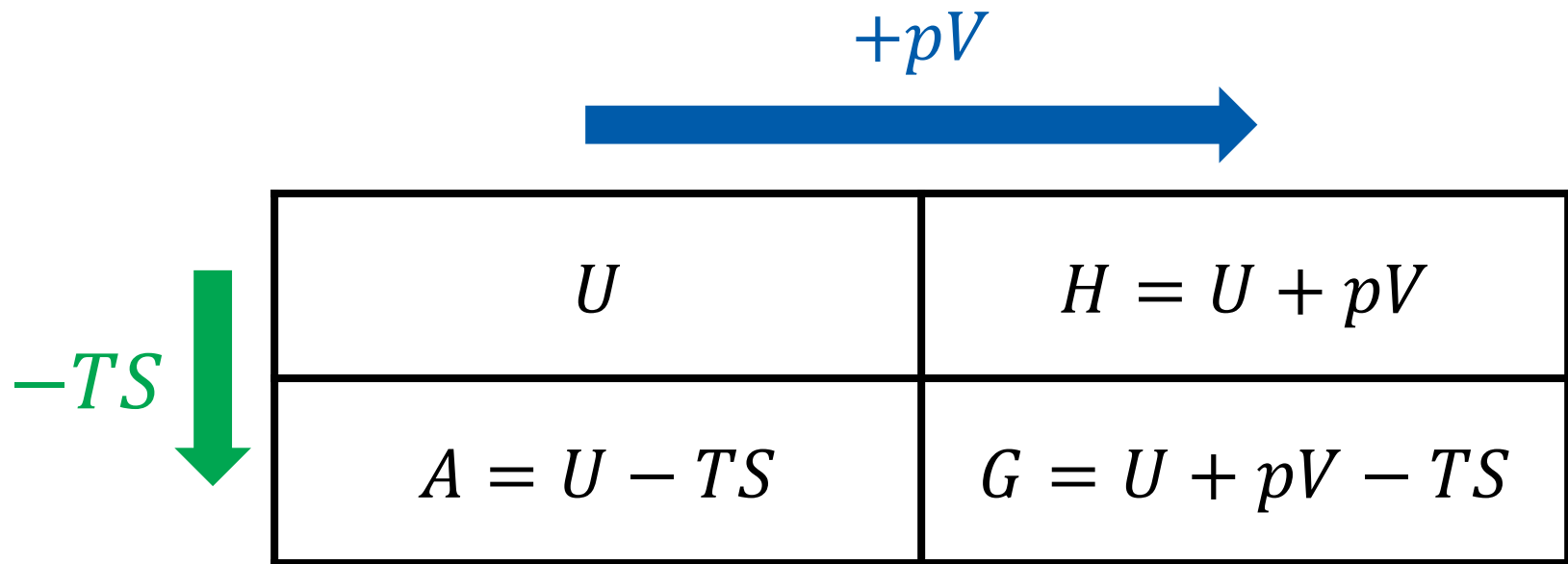
- $(S, V) \rightarrow (T, p)$

$$G = U + pV - TS \text{ (Gibbs energy)}$$

$$dG = dU + d(pV) - d(TS) = -SdT + Vdp$$

$$\therefore G = G(T, p)$$

Thermodynamic Potentials



Natural Independent Variables

- Internal energy $U = U(S, V)$

$$dU = TdS - pdV$$

- Enthalpy $H = U + pV = H(S, p)$

$$dH = TdS + Vdp$$

- Helmholtz energy $A = U - TS = A(T, V)$

$$dA = -SdT - pdV$$

- Gibbs energy $G = U + pV - TS = G(T, p)$

$$dG = -SdT + Vdp$$

Two Systems in Equilibrium

$$dU = TdS - pdV$$

- Mechanistic equilibrium: $p_1 = p_2$
 - Conjugate extensive variable: dV
- Thermal equilibrium: $T_1 = T_2$
 - Conjugate extensive variable: $dS = \delta q/T$
- Diffusive equilibrium: ?
 - Conjugate extensive variable: dn



Chemical Potential

- Diffusive equilibrium: $\mu_1 = \mu_2$
 - Conjugate extensive variable: dn
- (Modified) differentials of thermodynamic potentials

$$dU = TdS - pdV + \mu dn$$

$$dH = TdS + Vdp + \mu dn$$

$$dA = -SdT - pdV + \mu dn$$

$$dG = -SdT + Vdp + \mu dn$$

Three Independent Variables

$$dU = TdS - pdV + \mu dn$$

This implies that $U = U(S, V, n)$, which leads to

$$dU = \left(\frac{\partial U}{\partial S}\right)_{V,n} dS + \left(\frac{\partial U}{\partial V}\right)_{S,n} dV + \left(\frac{\partial U}{\partial n}\right)_{S,V} dn$$

Comparison gives

$$\left(\frac{\partial U}{\partial S}\right)_{V,n} = T, \quad \left(\frac{\partial U}{\partial V}\right)_{S,n} = -p, \quad \left(\frac{\partial U}{\partial n}\right)_{S,V} = \mu$$

Meaning of μ

$$dG = -SdT + Vdp + \mu dn$$

This implies that $G = G(T, p, n)$.

G is extensive, and n is the only extensive variable. Thus,

$$G(T, p, n) = ng(T, p)$$

and

$$\left(\frac{\partial G}{\partial n}\right)_{T,p} = \left[\frac{\partial\{ng(T, p)\}}{\partial n}\right]_{T,p} = g(T, p)$$

Meaning of μ

$$g(T, p) = \frac{G}{n} = \left(\frac{\partial G}{\partial n} \right)_{T, p}$$

However,

$$dG = -SdT + Vdp + \mu dn$$

gives

$$\mu = \left(\frac{\partial G}{\partial n} \right)_{T, p} = g(T, p) = \frac{G}{n}$$

(molar Gibbs energy)

Thermodynamic Potentials

$$\mu = G/n$$

can be written as

$$G = \mu n$$

From this equation, we can obtain

$$H = G + TS = \mu n + TS$$

$$A = G - pV = \mu n - pV$$

$$U = G + TS - pV = \mu n + TS - pV$$

Two Systems in Equilibrium

$$dU = TdS - pdV + \mu dn$$

- Mechanistic equilibrium: $p_1 = p_2$
 - Conjugate extensive variable: dV
- Thermal equilibrium: $T_1 = T_2$
 - Conjugate extensive variable: $dS = \delta q/T$
- Diffusive equilibrium: $\mu_1 = \mu_2$
 - Conjugate extensive variable: dn



Multi-Component System

- State: $(p, T, n) \rightarrow (p, T, n_A, n_B, \dots)$
- Thermodynamic potentials
 - $dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dG = -SdT + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$

Thermodynamics of Phases

At equilibrium, the **chemical potential** of a substance is the **same** in and throughout every phase in the system

